Preciting Code-Switching: Comparitive Analysis of Data

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I do not have an abstract yet.

1 Introduction

1.1 The Bilingual Experience

Bilingualism is a growing experience across the world, whether it is through late second-language acquisition, early childhood immersion, or other varied source of exposure in a globally connected stage of the world. Bilingualism, as it has been studied, is an ability that creates a cognitive, behavioral, and neural difference from a monolingual experience (). Second-language acquisition even at a perceived lower-than-mastery level can be linked to varying levels of advantages in executive functions, which includes working memory, inhibitory control, and flexibility (for a review see Bialystok & Poarch, 2014). The prevailing conclusion for these differences is that bilinguals have a distinct structure for language than monolinguals. That is to say, bilinguals have a different form of language production and comprehension.

1.2 Code-Switching

Even within bilingual communities further differences are observed in language control through code-switching. Language control allows for the delivery of the correct language in speech, preventing unintended insertions of a secondary language while talking to a speaker that only understands one. Alternatively, Code-switching is the structured use of alternating languages within a conversation. Code-switching is seen for single word insertion, sentence switch, or phrasal switches. The alternation between languages is seen to incur a switching cost which is a delayed response time between the previous and current language in face of inhibition demands. The patterns of code-switching are not well understood, but recent studies use linguistic signals to determine possible syntactic and grammatical restrictions imposed on code-switching languages. For example, combinations of determiner and nouns can be measured for congruency, in languages with morphologically coded agreement, and commonality of use for those phrases. Beatty-Martínez & Dussias (2017) measure these differences through the N400 effect, which is linked to sensibility to unexpected stimuli, for habitual code-switchers and non-codeswitchers. They found a greater N400 effect for incongruency for code-switchers and a general early frontal activity to code-switching in

non-codeswitchers. This study further reflects the theoretical model of Adaptive Control Hypothesis that poses a flexibility in language control depending on conversational contexts, signaling there is a degree of predictability in which code-switching is structured in relation to an individual's interaction with their language.

1.3 Prediction

Prediction is seen as the ability to anticipate the upcoming word by semantic constraint or context. Auditorably, phonetic variation, including voice onset and speech rate, occurs as a signal of code-switching aiding in the anticipation of a code-switch (Fricke et al., 2016). Other linguistic signals observed manifest at a syntactic level, through preference as seen in Beatty-Martínez & Dussias (2017). Fricke & Kootstra (2016) explored the factors that trigger code-switching as a language activation effect utilizing the Bangor Miami Corpus of English-Spanish codeswitching (Deuchar et al., 2014), where they found that the presence of a prior code-switch triggered following code-switches for close distance utterances and words. Valdés Kroff (2016) focused on code-switching between determiners and noun phrases in the Bangor Miami Corpus and found that in gendered determiners of Spanish there is a significant preference for masculine determiners regardless of the noun phrase's Spanish gender. Valdés Kroff concludes masculine determiners masculine determiners can function as a predictor for a code-switch of the noun phrase.

1.4 The Current Analysis

The current analysis aims to investigate the syntactic prediction brought by part of speech in bilingual conversation by comparing a bilingual corpus to experimental data on predictive code-switching. The first part of the analysis is a replication of prior work comparing use of determiners as markers of code-switches as a base of validity for experimental data analysis in corpus work. Then the analysis seeks to explore particular qualities pertaining to other variable effects on code-switching.

2 Methods

The first section of the study utilized the open-access Miami Bilingual Corpus (Deuchar et al. 2014). This data was compiled from conversational recordings of 84 participants living in Miami, Florida, which was then automatically glossed for part of speech and language. The speakers themselves are anonymous, with speech from non-consenting participants being redacted from the corpus.

3 Analysis

4 Conclusions